Cotton

Gossypium hirsutum — The White Gold of Pakistan's Economy

A Complete Cultivation Guide for

Punjab · Sindh · Khyber Pakhtunkhwa · Balochistan

Compiled for Farmer Guidance & AI Advisory Use

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1. Introduction

Cotton, often called "white gold," has historically been one of Pakistan's most economically important crops, forming the backbone of the country's large textile industry and a major source of export earnings. It is grown mainly as a summer (kharif) crop across Punjab and Sindh's irrigated plains, where it fits into rotation with wheat as one half of the country's dominant cotton-wheat cropping system. Cotton production in Pakistan has faced serious challenges in recent years from pest pressure, particularly whitefly and pink bollworm, along with climate-related stresses, making improved varieties and management practices especially important. This guide explains how cotton is grown across Pakistan's cotton belts and how yield and fiber quality can be improved.

2. Scientific Name

Cotton's scientific name is Gossypium hirsutum (upland cotton, the dominant type grown in Pakistan), belonging to the Malvaceae family. A small amount of Gossypium arboreum (desi cotton) is also grown in some areas for its hardiness, though it has largely been replaced by higher-yielding hirsutum varieties and hybrids.

3. Importance in Pakistan

Cotton is central to Pakistan's economy, feeding the country's large textile and garment export industry, which is one of the largest sources of foreign exchange earnings for the country. It is grown on roughly 2–2.5 million hectares, primarily in Punjab and Sindh, and supports the livelihoods of millions of farming households as well as workers throughout the ginning, spinning, and textile manufacturing value chain. Cotton seed is also crushed for cooking oil and its cake used as livestock feed, adding further value beyond the fiber itself, making the crop's health and productivity a matter of significant national economic concern.

4. Major Producing Provinces and Districts

- Punjab: Multan, Bahawalpur, Vehari, Rahim Yar Khan, Dera Ghazi Khan, and Muzaffargarh — the largest cotton-growing region in the country.
- Sindh: Sanghar, Nawabshah (Shaheed Benazirabad), Hyderabad, Mirpurkhas, and Ghotki — the second major cotton belt, contributing significantly to national production.
- Khyber Pakhtunkhwa and Balochistan: Limited cultivation, mainly in southern KP districts bordering Punjab's cotton belt, and in parts of Balochistan with reliable irrigation.

5. Climate Requirements

Cotton is a warm-season crop requiring temperatures between 25°C and 35°C during the growing season, with a long, frost-free growing period of at least 180 days needed for full maturity. It requires good sunlight, especially during flowering and boll development, and can tolerate moderate drought once established due to its relatively deep root system, but excessive rainfall or humidity during boll opening can cause boll rot and reduce fiber quality.

6. Soil Requirements

Cotton grows best in well-drained, deep loam to clay loam soils with good water-holding capacity, and a soil pH between 6.0 and 8.0, tolerating the mildly alkaline soils common across Punjab and Sindh's irrigated plains reasonably well. Waterlogging is harmful to cotton, particularly in the early growth stages, so fields prone to standing water should be properly leveled and drained before sowing.

7. Land Preparation

Land is typically ploughed two to three times followed by planking to prepare a fine, well-leveled seedbed, since good early root establishment supports the crop's long growing season. Farmyard manure, where available, is often incorporated ahead of sowing, and proper field leveling using laser leveling technology, where accessible, helps improve irrigation efficiency across the crop's long growing season.

8. Recommended Varieties (Pakistan)

- Bt cotton hybrids and varieties (genetically modified for resistance to bollworm pests) are now dominant across Punjab and Sindh, given their significant advantage in reducing bollworm-related losses and insecticide costs.
- Varieties developed and recommended by the Central Cotton Research Institute Multan, the Cotton Research Institute Faisalabad (Punjab), and the Central Cotton Research Institute Sakrand (Sindh) are regularly updated to address emerging pest and disease pressures, including whitefly-transmitted Cotton Leaf Curl Virus.
- Farmers are strongly advised to use certified seed from approved sources each season and to confirm the latest recommended varieties from provincial seed corporations and cotton research institutes, since resistance-breaking pest and disease strains periodically require new variety releases.

9. Seed Rate

Recommended seed rate for cotton is about 6–8 kg per acre for delinted (mechanically or acid-treated to remove fuzz) certified seed, while fuzzy (untreated) seed requires a higher rate of around 10–12 kg per acre due to less uniform planting and lower effective germination.

10. Seed Treatment

Seed should be treated with a fungicide such as carbendazim or thiram (2–3 g per kg seed) to protect against seedling diseases, and an insecticide seed treatment is often applied as well to protect young seedlings from early-season sucking pests such as jassids and whitefly, which is especially important given the crop's vulnerability to Cotton Leaf Curl Virus spread by whitefly at the seedling stage.

11. Sowing Time (Province-wise)

- Punjab: Sown mid-April to May in most areas, with early sowing generally preferred to allow the crop to establish before peak summer heat and to mature before the cooler, wetter conditions of late autumn that can affect boll opening and fiber quality.
- Sindh: Sown April–May, similar to Punjab, adjusted to local irrigation water availability.
- Khyber Pakhtunkhwa and Balochistan: Sown April–May in the limited areas where cotton is grown.

12. Sowing Methods

Cotton is generally sown by drilling seed in rows about 75 cm apart with plant-to-plant spacing of 20–30 cm, at a depth of about 4–5 cm, using a seed drill for uniform spacing where available. Proper plant population and spacing are important for balancing vegetative growth with boll development, since overcrowded plants tend to produce excessive vegetative growth at the expense of fruiting, while widely spaced plants may not fully utilize the land's yield potential.

13. Fertilizer Recommendations (NPK)

A general recommendation for cotton is 100–150 kg Nitrogen, 60–75 kg Phosphorus (P2O5), and 50–60 kg Potash (K2O) per acre, adjusted based on soil test results and expected yield potential. Full phosphorus and potash and about one-third of nitrogen are applied at sowing, with the remaining nitrogen split into two or three top-dressings at squaring (bud formation), flowering, and boll development stages, since excessive early nitrogen can promote excessive vegetative growth at the expense of fruiting and delay maturity.

14. Irrigation Schedule

Cotton requires irrigation approximately every 15–21 days depending on soil type and season, with critical irrigations needed at squaring, flowering, and boll development stages; water stress during these periods can cause flower and boll shedding, significantly reducing yield. Irrigation should be reduced as bolls begin opening to promote proper boll maturation and fiber quality, and excess late-season irrigation can delay maturity and increase disease risk.

15. Weed Management

Cotton's relatively slow early growth and wide row spacing make weed control important, especially in the first 45–60 days after sowing, typically managed through two to three hoeings or inter-row cultivation. Pre-emergence and post-emergence herbicides approved for cotton are used on many commercial farms to reduce labor requirements, particularly important given cotton's long growing season during which weeds can otherwise persistently compete for water and nutrients.

16. Insect Pests

Whitefly (the primary vector of Cotton Leaf Curl Virus and one of the most damaging pests overall), pink bollworm, American bollworm, jassids, thrips, and mealybug are the major insect pests affecting cotton in Pakistan, with whitefly and pink bollworm causing particularly severe economic losses in recent years despite widespread Bt cotton adoption. Integrated pest management combining resistant varieties, regular field scouting, pheromone traps for bollworm monitoring, conservation of natural predators, and need-based, rotated insecticide application is essential, since over-reliance on insecticides has led to resistance development in several key pest populations.

17. Diseases

Cotton Leaf Curl Virus (CLCuV), transmitted by whitefly, is the most economically devastating disease of cotton in Pakistan, capable of causing severe yield losses in susceptible varieties, and has driven much of the country's variety development efforts over the past two decades. Bacterial blight and Fusarium wilt can also cause significant damage in some areas. Management relies heavily on resistant varieties, whitefly control, removal of alternate host weeds, and crop rotation to reduce disease carryover between seasons.

18. Deficiency Symptoms

Nitrogen deficiency shows as pale yellow-green older leaves and reduced vegetative and fruiting growth. Phosphorus deficiency causes stunted growth, delayed flowering, and dark green or reddish-purple leaf discoloration. Potassium deficiency, a well-documented and yield-limiting problem in Pakistani cotton, appears as yellowing and browning (bronzing) along leaf margins, premature leaf shedding, and poor boll development, and is often under-corrected relative to nitrogen in farmer fertilizer practice. Magnesium and boron deficiencies can also occur and affect boll retention and fiber quality.

19. Harvesting

Cotton bolls are ready for picking when they fully open and the white lint is exposed and fluffy, with picking generally beginning around 150–160 days after sowing and continuing over multiple pickings (typically two to four) as bolls mature and open in succession over several weeks. Hand-picking remains the dominant harvesting method in Pakistan, valued for producing cleaner, less contaminated cotton compared to mechanical harvesting, which is important for maintaining fiber quality standards demanded by the textile industry.

20. Storage

Seed cotton (raw cotton with seed still attached, as picked from the field) should be stored in a dry, well-ventilated place and ginned (to separate lint from seed) as soon as practically possible to avoid quality deterioration and contamination. Proper storage and handling to avoid moisture, foreign matter contamination, and mixing of different varieties or grades is critical, since contamination and quality issues in raw cotton directly affect the price and export competitiveness of Pakistan's textile products.

21. Expected Yield in Pakistan

The national average cotton yield in Pakistan has fluctuated in recent years, generally in the range of 600–750 kg of lint per hectare, with significant year-to-year variation driven by pest pressure (particularly whitefly and pink bollworm), weather extremes, and input availability. Progressive farmers using resistant varieties, balanced fertilization (especially adequate potassium), and effective integrated pest management in Punjab and Sindh can achieve considerably higher yields than the national average.

22. Government Recommendations

The Central Cotton Research Institutes at Multan, Faisalabad, and Sakrand, along with provincial agriculture departments, lead variety development, pest monitoring, and extension advisory services for cotton, given the crop's critical importance to Pakistan's textile export economy. Government policy has periodically included support prices, certified seed subsidies, and coordinated pest management campaigns, particularly in response to major whitefly and Cotton Leaf Curl Virus outbreaks that have significantly affected national production in past seasons.

23. Modern Farming Practices

Bt cotton adoption for bollworm resistance, laser land leveling for irrigation efficiency, and increasing use of certified seed are among the most significant modern practice shifts in Pakistan's cotton sector over recent decades. Drip irrigation and precision fertilizer application are being adopted on some larger commercial farms to improve water and input use efficiency across the crop's long growing season.

24. Precision Agriculture

Precision approaches being introduced in cotton include soil testing to guide balanced NPK application, particularly to correct widespread potassium deficiency, and mobile-based pest monitoring and alert systems to help farmers time insecticide applications more effectively against whitefly and bollworm. Satellite and drone-based crop monitoring are being piloted by some research institutes and larger farms to detect pest and disease hotspots and irrigation needs earlier across large cotton-growing tracts.

25. Sustainable Farming

Sustainable cotton practices being promoted in Pakistan include integrated pest management to reduce over-reliance on insecticides and slow resistance development in whitefly and bollworm populations, balanced fertilization guided by soil testing rather than routine nitrogen-heavy application, and crop rotation with wheat and other crops to break pest and disease cycles associated with continuous cotton cultivation. Efficient irrigation methods, including drip irrigation where feasible, are also being encouraged to reduce water use in this water-intensive crop.

26. Regional Notes

Punjab's cotton belt, especially around Multan and Bahawalpur, and Sindh's cotton-growing districts together form the backbone of Pakistan's textile industry, though both regions continue to grapple with the persistent challenges of whitefly, Cotton Leaf Curl Virus, and pink bollworm that have affected national production in recent years. Continued investment in resistant varieties, integrated pest management, and balanced fertilization, particularly adequate potassium, remains essential for restoring and sustaining the productivity of this historically vital crop for Pakistan's economy.